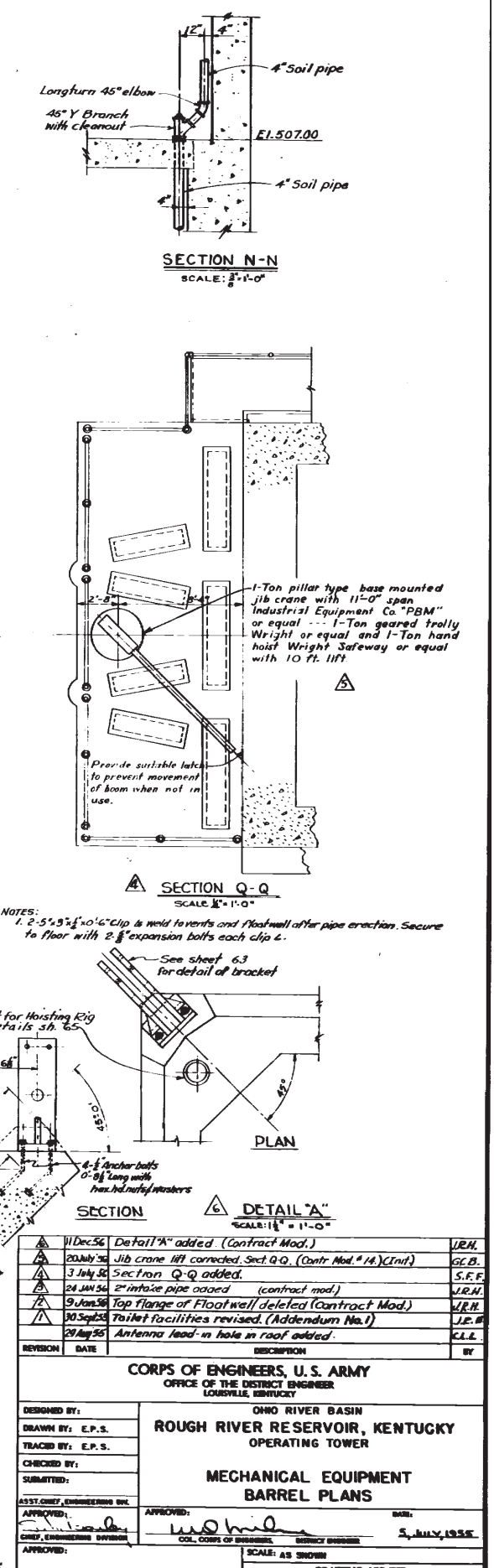
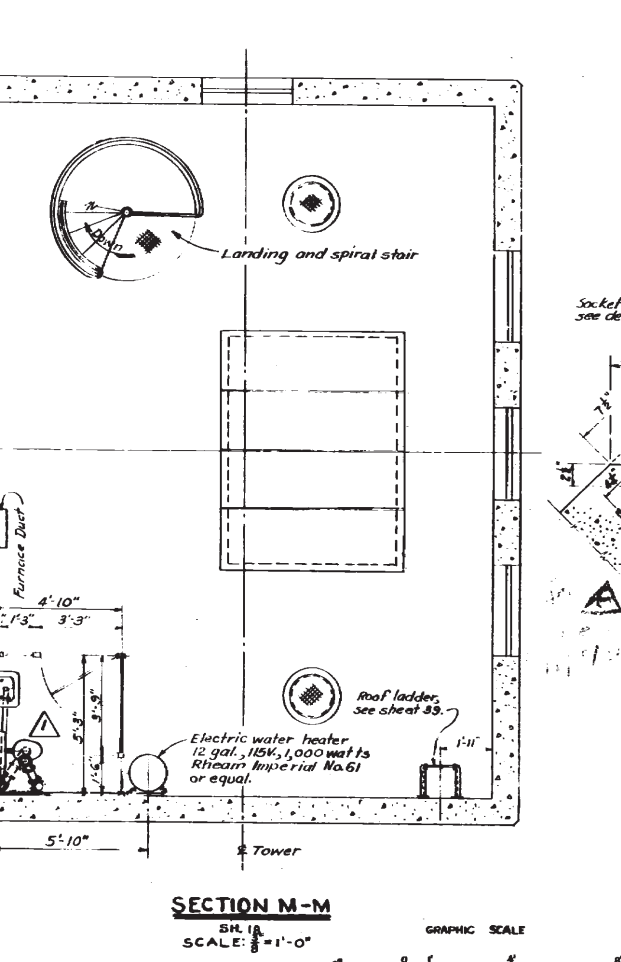
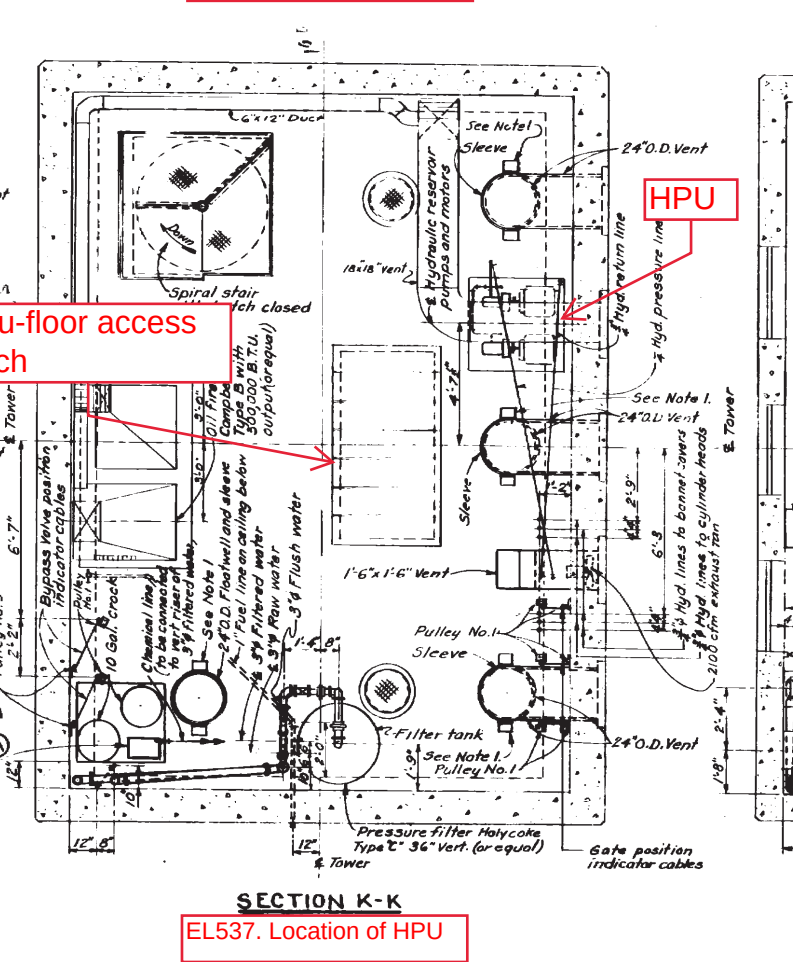
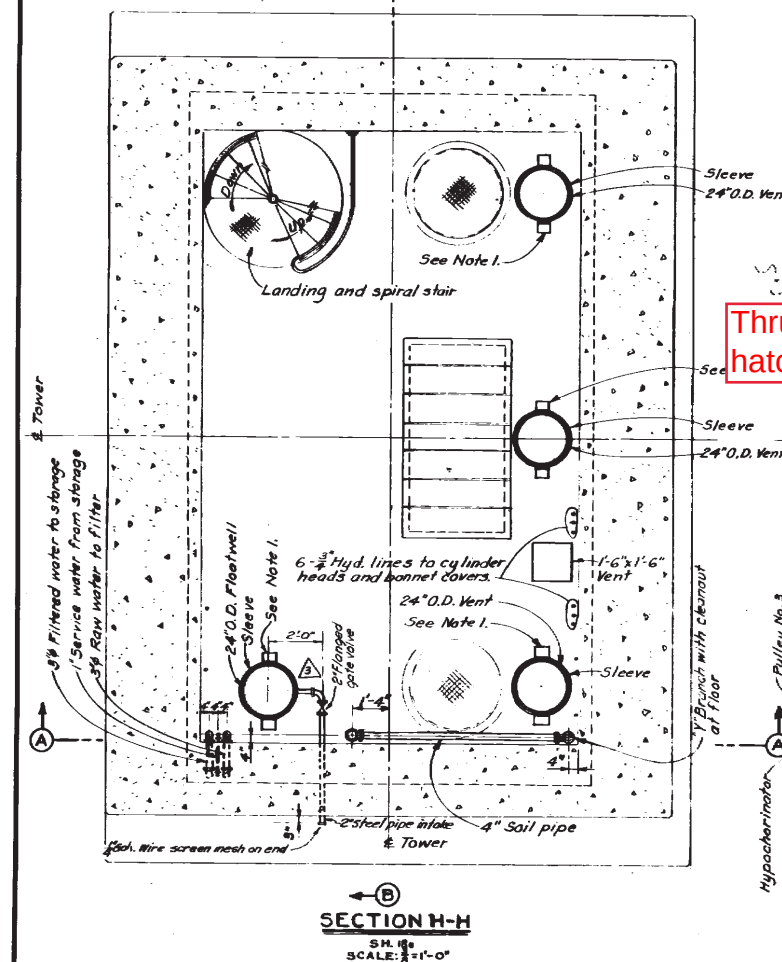
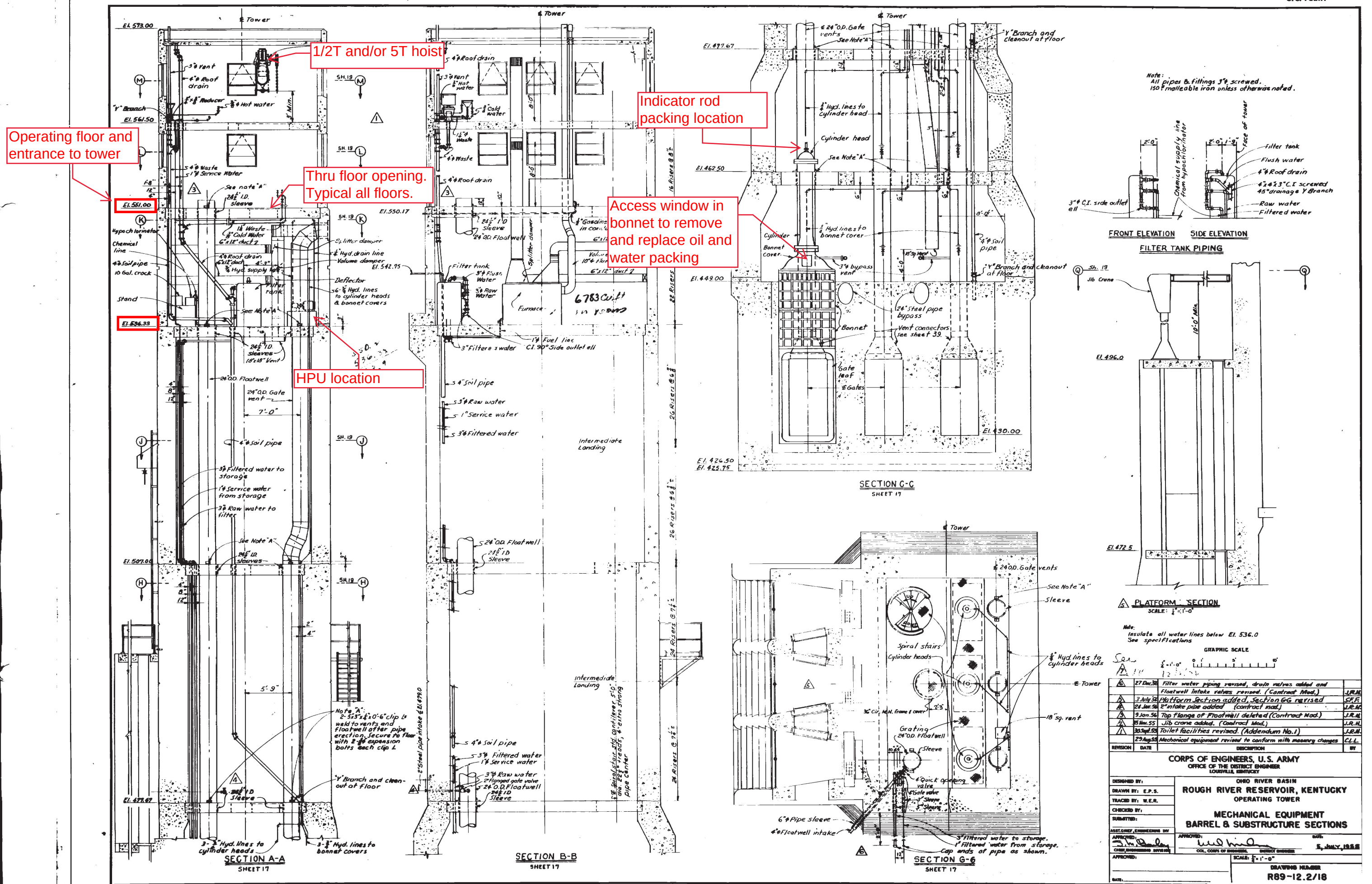


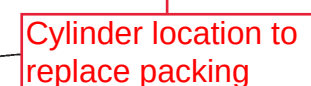


Δ	11 Dec 56	Data 1/A added. (Contract Mod.)	J.R.N.
Δ	20 July 56	Jib crane lift corrected. Sect Q-Q. (Contr Mod. # 1.) (Trut)	G.E.B.
Δ	3 July 56	Sect Fron Q-Q added.	S.E.F.
Δ	24 Jan 56	2" intake pipe added (contract mod.)	J.R.N.
Δ	9 Jan 56	Top Flange of Float well deleted (Contract Mod.)	J.R.N.
Δ	30 Sept 55	Toilet facilities revised. (Addendum No.1)	J.R.N.
Δ	29 Aug 55	Antenna lead-in hole in roof added.	C.L.E.
REVISION	DATE	DESCRIPTION	BY

CORPS OF ENGINEERS, U. S. ARMY
OFFICE OF THE DISTRICT ENGINEER
LOUISVILLE, KENTUCKY

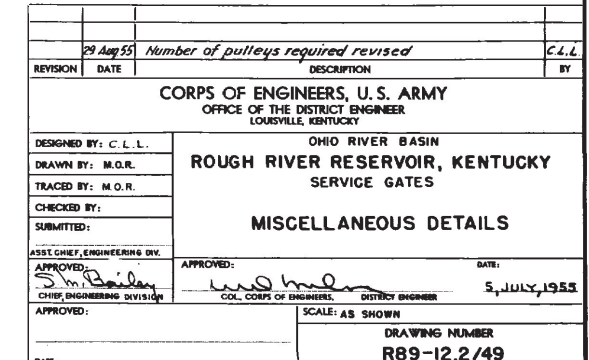
DESIGNED BY:	OHIO RIVER BASIN
DRAWN BY: E.P.S.	ROUGH RIVER RESERVOIR, KENTUCKY
TRACED BY: E.S.S.	OPERATING TOWER
CHECKED BY:	
SUBMITTED:	
ASST. CHIEF, ENGINEERING DIV.	
APPROVED:	
CHIEF, ENGINEERING DIVISION	
APPROVED:	
CHIEF, CORPS OF ENGINEERS	
SCALE: AS SHOWN	
DRAWING NUMBER	
R89-122-119	

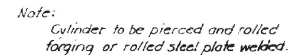




23 Sept 56	Filler water piping revised. Drain valves added. (Contr. Mod.)	J.B.H.
20 July 56	Service & filtered water lines revised. (Contr. Mod. #9)	G.C.B.
17 July 56	W/ve Oper. wrench deleted. Platform revised. Sect. A-A	S.F.F.
2 April 56	Rear Water Intake revised and details added. (Contr. Mod.)	P.W.B.
24 Jan 56	2" intake pipe added. (Contract mod.)	J.P.H.
30 Nov 55	Sheet reference added. (Add. No. 3 19 Oct. 55)	G.C.B.
15 Nov 55	Jib crane added. (Contract Mod.)	J.R.M.
29 Aug 55	Mechanical equipment revised to conform with masonry changes.	C.L.L.
REVISION	DATE	DESCRIPTION
BY		
CORPS OF ENGINEERS, U.S. ARMY OFFICE OF THE DISTRICT ENGINEER LOUISVILLE, KENTUCKY		
DESIGNED BY:	OHIO RIVER BASIN	
DRAWN BY: E.P.S.	ROUGH RIVER RESERVOIR, KENTUCKY	
TRACED BY: E.P.S.	OPERATING TOWER	
CHECKED BY:	MECHANICAL EQUIPMENT	
SUBMITTED:	SUBSTRUCTURE SECTIONS	
ASST. CHIEF ENGINEERING DIV.	APPROVED:	DATE:
<i>[Signature]</i>	<i>[Signature]</i>	5 JULY 1956
CHIEF ENGINEERING DIVISION	COL. CORPS OF ENGINEERS	DISTRICT ENGINEER
APPROVED:	SCALE: AS SHOWN	
DATE:	DRAWING NUMBER	
	R89-12.2/17	

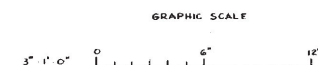
Oil and Water Packing





MARK 3 MARK 48/1
MATERIAL: FORGED STEEL ASTM A235-54, CLASS C
OR
STEEL PLATE ASTM A235-54, GR C
AND FORGED STEEL FLANGES ASTM A105-46, GR 1
SCALE: 3/4"=1'-0"

CYLINDER ASSEMBLY



REVISION	DATE	DESCRIPTION	BY
CORPS OF ENGINEERS, U.S. ARMY OFFICE OF THE DISTRICT ENGINEER LOUISVILLE, KENTUCKY			
DESIGNED BY: C.L.L.	OHIO RIVER BASIN ROUGH RIVER RESERVOIR, KENTUCKY SERVICE GATES CYLINDER ASSEMBLY		
DRAWN BY: W.J.B.			
TRACED BY: W.J.B.			
CHECKED BY:			
SUBMITTED:			
ASST. CHIEF, ENGINEERING DIV.			
APPROVED:	APPROVED:	DATE:	
<i>Chas. G. Hef</i>	<i>W. J. B.</i>	5 JULY 1955	
CHIEF ENGINEERING DIVISION	COL, CORPS OF ENGINEERS, DISTRICT ENGINEER		
APPROVED:	SCALE: AS SHOWN		
DRAWING NUMBER R89-12.2/48			
DATE:			

PARTS FOR HYDRAULIC OIL PRESSURE SYSTEM

MARK	NOMENCLATURE	SIZE	WORKING PRESSURE	REMARKS
B	Bushing	3/4" x 1/2"	1000*	
CC	Cap - Oil Filler & Air Filter Unit			Vickers SP-113-B or equal.
E	Elbow - 90°	3/4"	1000*	
E1	Elbow - 90° - Street	3/4"	1000*	
E2	Elbow - 90° - Female Union	1/2"	150*	
E3	Elbow - 90° - Male Union	1/2"	150*	
F	Flange - Tank with gasket	3/4"		Vickers 33392 or equal.
G	Gage - Pressure - Bourdon Type	6" Dia.		Calibrated to 1200*
G1	Gage - Liquid level - bronze & ball checks in gage valves			Pemberly Injector Co. Safeguard Auto. Liquid Level Gage No. 50 or equal.
N	Nipple - Extra strong	3/4"	1000*	
P	Pipe - Steel - Extra strong	3/4"	1000*	
P1	Pipe - Steel - Standard	1/2"	150*	
R	Retainer	1/2"		Vickers 35092 or equal.
S	Strainer, Oil Filter Element	1/2"		Vickers 29092 or equal.
T	Tee	3/4"	1000*	
U	Union - Female	3/4"	1000*	
U1	Union - Female	3/4"	1000*	
V	Valve - Gate	3/4"	1000*	
V1	Valve - Relief	1000*		Integral with pump unit
V2	Valve - Check - Right Angle	1000*		Vickers C-815 or equal
V3	Valve - Air Release	1000*		Logan 9040 or equal
V4	Valve - 4 Way - 3 Position Detents	1000*		Racine Bulletin 34 or equal
V5	Valve - Angle - Hy - Pressure - Steel	1000*		Crane 223 H or equal
V6	Valve - Counter Balance	1000*		Vickers RCT-06-D110
V7	Valve - Angle - Hy - Pressure - Steel	1000*		Crane 223 H or equal
C1	Cross tube to tube to tube	3/4" a.d.		Imperial brass 752-FS or equal
C2	Connector 3/4" a.d. tube to 1/2" M.P.T.			Imperial brass 768-FS or equal
C3	Connector 3/4" a.d. tube to 1/2" M.P.T.			Imperial brass 768-FS or equal
N1	Nipple - Std. Str.	1/2"	150*	
C	Coupling, Pipe	1/2"	150*	
F1	Flange - Tank with gasket	3/4"		Vickers 33391 or Equal
P1	Tubing	3/4" a.d.		

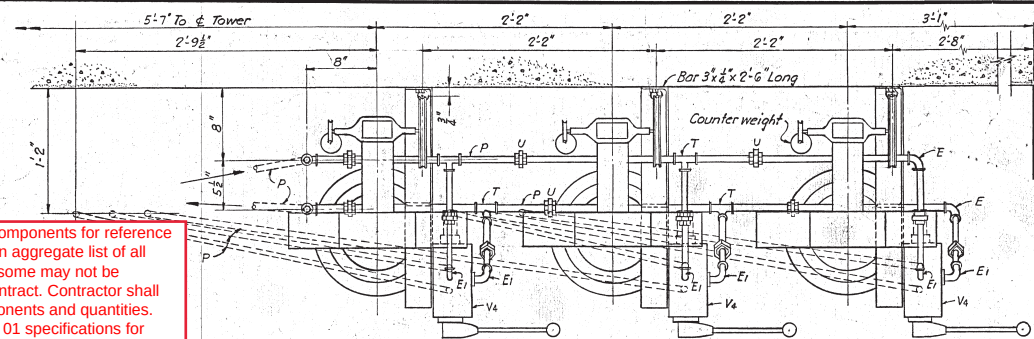
Schedule of components for reference only. This is an aggregate list of all components, some may not be included in contract. Contractor shall confirm components and quantities. See Line item 01 specifications for components with updated requirements.

Approximate location of pressure gauge install, line item 04

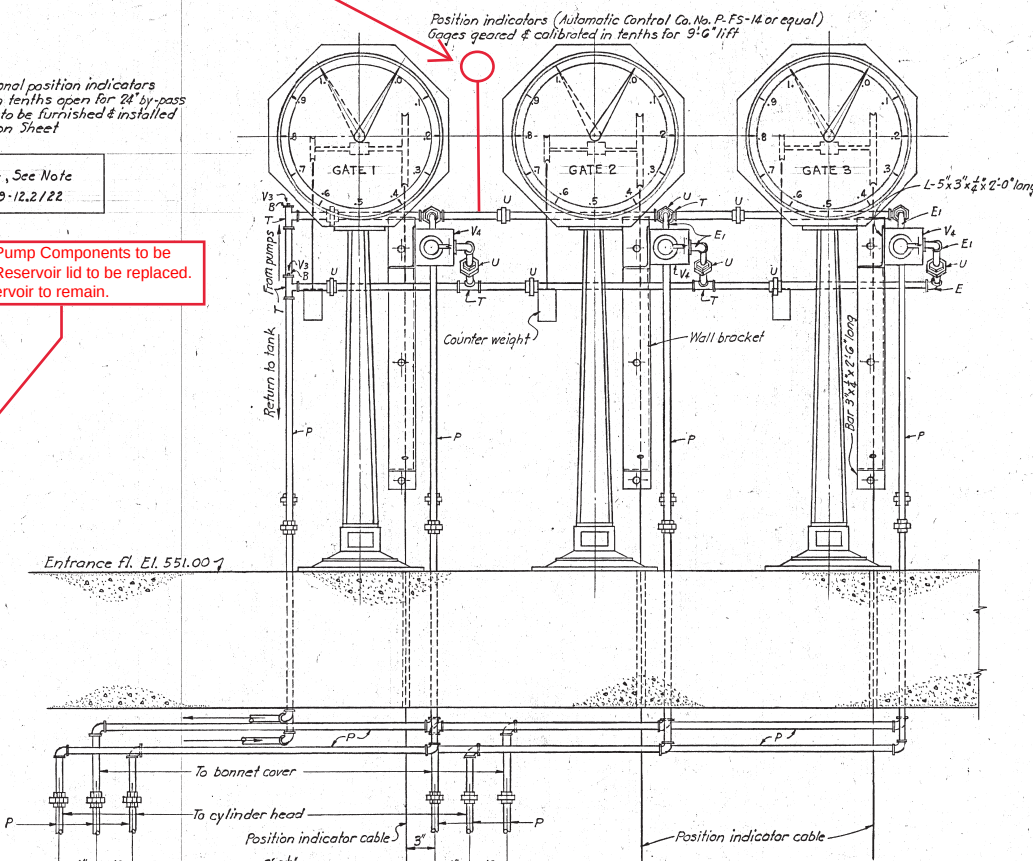
Note: 2-Additional position indicators calibrated in tenths open for 24" by-pass gate valves to be furnished & installed as located on sheet

Modify, See Note SH-R89-12.2/22

Hydraulic Pump Components to be replaced. Reservoir lid to be replaced. rest of reservoir to remain.



PLAN

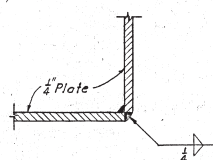


ELEVATION

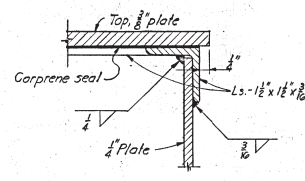
VALVES & INDICATORS

ENTRANCE FLOOR

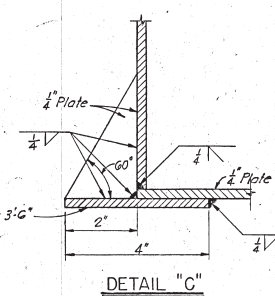
SCALE: 1 1/2" = 1'-0"



DETAIL "A"



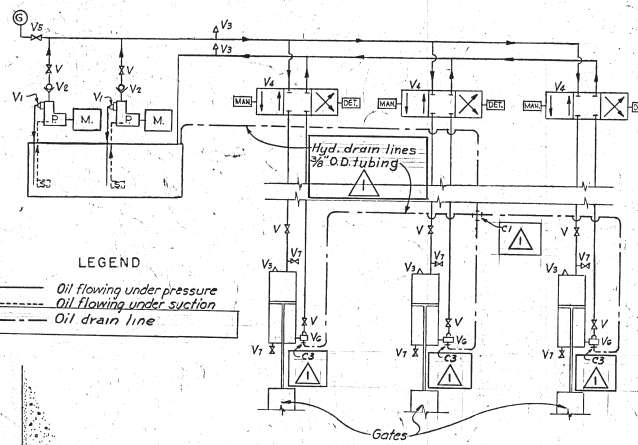
DETAIL "B"



DETAIL "C"

GRAPHIC SCALE

1 1/2" = 1'-0"



SCHEMATIC PIPING DIAGRAM

SCALE: NONE

NOTE
WORK TO BE DONE UNDER THIS CONTRACT CONSISTS OF THAT INDICATED BY REVISION 1 (A) ONLY. ALL OTHER WORK WAS COMPLETED UNDER PREVIOUS CONTRACT.

DESIGNED BY: FL		DATE: 12 Sept 58		Flange & Fittings added to tank. Oil drain lines added to schematic piping diagram		BY: JMH	
DRAWN BY: W. J. B.		DATE:		DESCRIPTION:			
TRACED BY: W. J. B.		DATE:		CORPS OF ENGINEERS, U.S. ARMY			
CHECKED BY: J. R. H.		DATE:		OFFICE OF THE DISTRICT ENGINEER			
SUBMITTED BY: J. R. H.		DATE:		LOUISVILLE, KENTUCKY			
APPROVED BY: J. R. H.		DATE:		OHIO RIVER BASIN			
APPROVED BY: J. R. H.		DATE:		ROUGH RIVER RESERVOIR, KENTUCKY			
APPROVED BY: J. R. H.		DATE:		OPERATING TOWER			
APPROVED BY: J. R. H.		DATE:		HYDRAULIC OIL PRESSURE SYSTEM			
APPROVED BY: J. R. H.		DATE:		AND GATE POSITION INDICATOR			
APPROVED BY: J. R. H.		DATE:		SCALE: 1 1/2" = 1'-0"			
APPROVED BY: J. R. H.		DATE:		DRAWING NUMBER			
APPROVED BY: J. R. H.		DATE:		R89-12.2/61			
APPROVED BY: J. R. H.		DATE:		RECORD DRAWING - "AS BUILT"			

RECORD DRAWING - "AS BUILT"